

Jack Nystrom

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Education

University of California, Santa Cruz

Fall 2024 - Fall 2028 (Projected)

Computer Science B.S. and Electrical Engineering Minor

GPA: 3.88

Coursework in Embedded Programming, C and Assembly, Circuit Analysis, Physics, Analog Electronics

Experience

Formula Slug FSAE Team — **Electrical Captain**

June 2026 – Present

- Lead design of racecar electrical system including HV Battery, BMS, VCU, Powertrain, and all PCBs
- Teach ~20 members circuit design, schematic layout & routing in KiCad, part sourcing, and soldering
- Led testing and iteration of our first custom HV Battery pack, yielding 10% energy increase while 6 lbs lighter; fixed mistakes in cell balancing and isoSPI circuits, redesigning PCB within 2 hours
- Debug analog and digital waveforms with oscilloscopes, testing PCBs and shutdown circuit behavior

Formula Slug FSAE Team — **Firmware Lead**

Sept 2024 – June 2026

- Wrote STM32 firmware for ECUs including battery management, throttle control, data logging, and UI
- Fixed throttle control bugs at competition enabling us to pass active inspection for the first time
- Developed a traction control algorithm improving lap times using IMU, motor controller, and wheel speed data to prevent tire slip and ensure racecar always has traction
- Debugged and eliminated firmware crashes, enforcing static memory & non-blocking I/O in all firmware
- Achieved perfect score in LV & Data Acquisition for in-house telemetry system and harness reliability

Rivian VW Technologies — **SWE Intern, System Integration**

June 2026 – Sept 2026

- Created an Automotive Ethernet Gateway using embedded Linux to block and manipulate Zonal ECU messages, enabling automated functional safety testing on Rivian LabCars
- Designed kernel-level C networking stack, optimizing packet parsing to $<1\mu\text{s}$ using eBPF XDP
- Integrated Gateway onto Rivian R2 labcars, debugging Automotive Ethernet and CAN with Wireshark

Zero Motorcycles — **Software Intern**

June 2025 – Jan 2026

- Developed motorcycle debugging tool for dealership technicians, enabling successful 1.0 release
- Improve reliability and implement new features in CAN communication, ECU firmware updates, cloud integration, and Flutter UI

Projects

FSAE Telemetry PCB

Nov 2024 – June 2025

- Created Telemetry data acquisition PCB using KiCad with an AVR ATTiny microcontroller
- Implemented wheatstone bridge and frequency-to-voltage converter for strain gauge and wheel speed
- Developed a baremetal C driver for the SPI-based MCP2515 CAN controller

FSAE Data Analysis Stack ([link](#))

Sept 2024 – Present

- Developed telemetry and data visualization systems, providing instant and accessible data insights
- Created full-stack data pipeline from C++ logging firmware to high-performance web visualizations

Skills and Interests

PCB Design: Digital & analog circuit design, board layout, KiCad/Altium

Embedded Software: STM32, Arm Cortex-M, AVR, RPi; baremetal, FreeRTOS, MbedOS, Embassy-rs.

Programming Languages: C, C++, Arm32 Assembly, Rust, Zig, Python, Bash, Lua, Java, JS/TS, Nix

Automotive & Robotics: CAN, DBC, Ethernet, SPI/I2C/UART/USB, PID control, Wire harnessing

Tools: Oscilloscope, Multimeter, Waveform generator, Logic analyzer, Power supplies, THT&SMT Soldering

Unix: Linux, Git, Systemd, NixOS, Kernel modules, GDB, Docker, CLI tools, TCP/IP